

Data Quality Contract

Retail Orders Dataset | retail-orders-raw.csv | Version 1.0 | Effective Date: 2026-09-12

1. Purpose

This contract translates the business request “make sure our retail order data can be trusted for revenue and operations reporting” into measurable, testable definitions. It binds the data-producing and data-consuming teams to a shared standard for what “trustworthy data” means, how it is measured, and what happens when it fails.

2. Roles & Ownership

Role	Responsibility
Decision Owner	Head of Growth & Revenue Operations — accountable for decisions built on this data (pricing, discounting, segment strategy) and final sign-off on this contract.
Data Producer	Checkout / Order Management System engineering team — owns the upstream source of order_id, order_date, quantity, unit_price, discount_pct, payment_status.
Data Steward	Data Engineer — runs the profiling checks on each refresh, maintains the quarantine queue, and triggers escalation.
Data Consumers	Revenue Analyst, Finance Manager, Marketing Manager, Product Manager, Inventory/Catalog Manager, Ops Manager (see KPI Dictionary for per-KPI ownership).

3. Quality Dimensions, Thresholds & Executable Checks

Each dimension below is measured by the executable checks in **retail_data_profile.ipynb**. A dimension is “in contract” only when its measured result meets or exceeds its threshold.

Dimension	Definition	Executable Check	Pass Threshold
Completeness	Required fields (order_id, order_date, city, quantity, unit_price, payment_status) are populated.	% non-null per required column, computed every refresh.	≥ 99% non-null per field
Uniqueness	Every order is represented exactly once.	COUNT of duplicate order_id values.	0 duplicates
Validity	Values conform to the type, range, and domain rules in the data dictionary (dates, positive quantity, 0–100% discount, allowed categories).	Per-column type/range/domain parsing; invalid rows listed by order_id.	≥ 99% valid on date/quantity/discount; 100% on category domain
Consistency	Categorical fields normalize to one canonical value set regardless of source casing/whitespace.	Case/whitespace-normalize customer_segment and payment_status, then check against the allowed set.	100% map to the canonical set after normalization
Freshness	Order data is available for reporting soon after it is placed.	Current time minus MAX(valid order_date) at each refresh.	≤ 24 hours lag at the daily refresh checkpoint

4. Baseline Measurement (retail-orders-raw.csv, 12-row sample)

The table below is the result of running the executable checks against the current raw file. It establishes the starting baseline this contract is meant to improve — it is **not** itself a passing state.

Dimension	Result	Status
Completeness	91.7% non-null on order_date, city, discount_pct	FAIL
Uniqueness	1 duplicate order_id (RT-1004)	FAIL
Validity — order_date	75.0% valid (9 of 12 rows)	FAIL
Validity — quantity	83.3% valid (10 of 12 rows)	FAIL
Validity — discount_pct	1 of 11 non-null values out of range	FAIL
Validity — category	100% valid	PASS
Consistency	100% after case normalization	PASS*
Freshness	Lag far exceeds 24h on this static sample	Not applicable to static sample

* Passes only once the normalization step in Section 3 is applied; raw values fail on case-sensitivity alone (e.g. “student” vs “Student”, “paid” vs “Paid”).

5. Failure Thresholds & Escalation Actions

Escalation is tiered by how far a dimension's measured result falls below its threshold, and by how many downstream KPIs the failure touches.

Severity	Trigger Condition	Escalation Action	SLA
Watch	Any dimension within 1–3 percentage points of its threshold.	Data Steward logs the drift in the profiling run notes. No page/alert sent.	Reviewed at next weekly data-quality standup
Warning	One dimension fails threshold, affecting ≤ 2 KPIs (e.g., an isolated Validity failure on discount_pct).	Data Steward notifies the Data Producer and the affected KPI Owner(s) via Slack/email within the SLA; offending rows are quarantined (excluded from KPI calculation, not deleted) pending correction.	Notify within 4 business hours of detection
Critical	Uniqueness fails (any duplicate order_id), OR Completeness/Validity fails on a required field used by Revenue KPIs (order_date, quantity, unit_price, payment_status), OR ≥ 2 dimensions fail simultaneously.	Data Steward halts the automated KPI refresh, escalates directly to the Decision Owner, and opens an incident ticket with the Data Producer. Revenue and Ops dashboards are flagged "Data Quality Hold" until re-validated.	Notify within 1 business hour; root-cause plan within 24 hours
Critical	Freshness lag exceeds 24 hours at the scheduled daily refresh.	Data Steward pages the pipeline on-call engineer; Decision Owner is notified that KPIs are stale and should not be used for same-day decisions.	Page within 30 minutes of missed refresh

6. Row-Level Remediation Rules

Applied automatically by the profiling notebook before any KPI is calculated:

- Exact duplicate order_id rows — keep the first occurrence, drop the rest, log the count.
- Missing order_date, city, quantity, or unit_price — quarantine the row (exclude from KPIs) and route to the Data Producer for correction at source; never impute a required field silently.
- order_date not in ISO format or outside 2025-01-01–today — quarantine and request source correction; do not guess the intended date.
- quantity ≤ 0 or non-numeric — quarantine; never coerce to a default quantity.
- discount_pct outside 0–100 — quarantine and flag to the Marketing/Pricing owner; missing discount_pct is treated as 0 only when the source system confirms no discount was offered.
- customer_segment / payment_status casing or whitespace variants — normalize automatically (Title Case, trimmed) rather than quarantining; this is a Consistency fix, not a data-loss event.

7. Review & Change Control

This contract is reviewed monthly by the Decision Owner and Data Steward, and immediately after any Critical escalation. Threshold changes require sign-off from both the Decision Owner and the Data Producer, and must be reflected in the KPI Dictionary's Filters/Exclusions column where relevant.

8. Sign-off

Decision Owner Signature: _____ Date: _____

Data Producer Signature: _____ Date: _____

Data Steward Signature: _____ Date: _____

Companion files: retail_kpi_and_quality_workbook.xlsx (KPI Dictionary, Data Profile Summary, Issue Log) and retail_data_profile.ipynb (executable checks that produce the figures in Sections 4 and 6).